



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

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Course Name: B.Tech.

Subject Name: Fundamentals of Linear Algebra and Statistics

Subject Code: 25AS102

Max. Marks: 15

Time: 75 Minutes

Semester: 1st

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Find the sum and product of the eigen values of the matrix $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	CO3	K2	2.1.3
(b)	State Cayley Hamilton Theorem.	CO3	K1	2.1.3
(c)	Determine 'b' such that the system of homogeneous equations $2x + y + 2z = 0, x + y + 3z = 0, 4x + 3y + bz = 0$ has a nontrivial solution.	CO3	K2	2.1.3
(d)	The set of all second degree polynomial is not a vector space under the vector addition and scalar multiplication over R. Why?	CO1	K2	2.4.2
(e)	Define Subspace.	CO1	K1	2.4.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Reduce the matrix A to normal form and hence find its rank, where $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$	CO3	K3	2.1.3
(b)	Employing elementary transformation find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$	CO3	K2	2.1.3
I	For what values of 'a' and 'b' the following system of linear equations: $x + 2y + 3z = 6, x + 3y + 5z = 9, 2x + 5y + az = b$, have (i) No solution (ii) Unique solution (iii) Infinitely many solutions	CO3	K3	2.1.3 5.2.2

SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question			(Same K Levels) (3 x 1)	
(a)	Find the eigen values and the corresponding eigen vectors of $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$.	CO3	K3	2.1.3
(b)	Verify Cayley Hamilton Theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .	CO3	K3	2.1.3
4. Attempt Any one Question			(Same K Levels) (3 x 1)	
(a)	Let $V = R^3$ be a vector space and $W = \{(x, y, z): x - 3y + 4z = 0\}$. Prove that W is a subspace of V over R.	CO1	K3	2.4.2 2.1.3
(b)	Prove that the set of all vectors in a plane (R^2) over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.	CO1	K3	2.4.1 2.4.4

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create